

</> Asadullah Shafique

Agentic AI Engineer | Multi-Agent Systems Architect | Enterprise AI Strategist

Bridging AI potential with financial services reality through orchestration, compliance, and scalable adoption.

📍 Karachi, Pakistan 📞 +92-321-3771445 ✉️ asadullahshafique@hotmail.com

🔗 github.com/asadullah48 🌐 LinkedIn 🌐 Portfolio Medium

📁 Professional Summary

Agentic AI Engineer building multi-agent systems for regulated industries, backed by **25+ years** of operating a manufacturing business. Most agentic pilots in financial services die at the compliance review — not because the models are weak, but because a system that cannot show *why* it reached a conclusion cannot be signed off by a second line of defence. My work makes that reviewable: typed orchestration with an auditable route, a written constitution with a deterministic layer that needs no model, and evaluation that reads the execution trace rather than trusting the prose.

Founder of Texcot Embroidery Sourcing House and creator of the **Textile ERP Platform**, targeting fabric mills, CMT units and garment exporters across Faisalabad, Sialkot, Gujranwala, Karachi and Lahore — so the domain intelligence is built from inside an industry rather than modelled from outside one. Completed **6 consecutive Panaversity Hackathons** (Bronze → Silver → Silver → Gold → Platinum → Agent Factory), each build **carrying the last one forward**.

</> Agentic AI Mastery

Twelve competencies, four categories. Each cites the path that demonstrates it.

AGENTIC ORCHESTRATION

تنسيق الوكلاء

- › Engineering Multi-Agent Ecosystem Design (MAS)
backend/orchestration/
- › Designing Graph-Based Agent Workflows
LangGraph · Plan-Act-Verify
- › Implementing Agentic Coding & SDLC Standardization
CLAUDE.md · spec-first

CONNECTIVITY & PROTOCOLS

الاتصال والبروتوكولات

- › Engineering MCP Interoperability Layers
/mcp/server · FastMCP
- › Standardizing Read-Only Agent Tool Contracts
6 tools, read-only
- › Architecting Provider-Agnostic Fallback Ladders
SDK → LangGraph → static

ENTERPRISE & SCALE

المؤسسات والتوسع

- › Engineering Domain-Specific Intelligence (Finance)
FinAgent-Nexus · 94 tests
- › Architecting Domain-Specific ERP Intelligence (Textile)
CMT · live
- › Implementing Cloud-Native Scale-Out
k8s/ · 11 manifests

RELIABILITY & SAFETY

الموثوقية والأمان

- › Implementing Deterministic Guardrails
constitution/ · 4-of-4 blocked
- › Designing Feedback-Driven Evaluation Loops
evals/ · 3 suites
- › Engineering Context Integrity & Trace Auditability
orchestration/context.py

</> Technical Skills

LANGUAGES

TypeScript

JavaScript

Python

HTML/CSS

BACKEND

FastAPI

SQLAlchemy ORM

PostgreSQL

REST APIs

CLOUD & DEVOPS

Kubernetes

Docker

Dapr

Kafka

Helm

GitHub Actions

Vercel

Koyeb

Minikube

PLATFORMS & TOOLS

Git

Windows 11

WSL2

FRONTEND

Next.js 15

React 19

Tailwind CSS v4

shadcn/ui

AI & AGENTS

OpenAI Agents SDK

MCP Servers

Constitutional AI

RAG

OpenAI API

OBSERVABILITY

Prometheus

Grafana

Jaeger

Loki

Hackathon Achievements

Panaversity Hackathon Series: 6 consecutive builds · Bronze → Agent Factory



Platinum 2025

Agent Factory (H5)

General Agent (Claude Code) manufactures Custom Agents (OpenAI Agents SDK). SKILL.md monetization. Digital FTE deployed on Kubernetes + Dapr. 117 - slide deck.

Claude Code

OpenAI SDK

MCP

SKILL.md

Dapr

K8s



Platinum 2025

Cloud-Native Deployment (H4)

Full K8s stack: Kubernetes + Docker + Dapr + Kafka + Prometheus/Grafana/Jaeger + Helm + GitHub Actions CI/CD.

K8s

Docker

Dapr

Kafka

Helm

Prometheus



Gold 2025

Advanced Todo (H3)

149 tests passing. Recurring todos, team collaboration, AI suggestions, triple-layer Constitutional AI (7 BLOCK + 5 FLAG patterns).

Next.js

FastAPI

PostgreSQL



Silver 2025

AI-Powered Todo (H2)

89 tests passing. Full-stack Constitutional AI todo. 70% code reuse from H1.

Next.js

TypeScript

FastAPI



Silver 2024

Course Companion FTE (H1)

Zero-Backend-LLM architecture. Digital FTE model. 70% code reuse from H0.

Next.js

TypeScript

OpenAI



Bronze 2024

Personal AI CTO (H0)

Flagship Projects

The Autonomous Agent Ecosystem

Shipped 

21 open-source platforms, one per problem domain

21 multi-agent platforms, each with its own FastAPI gateway, Helm chart, test suite and public repository — reference implementations running on synthetic domain data. 736 test functions across the 21 repositories, counted from source on 2026-09-10; 6 of them run those tests in CI. All open source.

FastAPI

Pytest

Helm

FinAgent - Nexus

Shipped 

Multi-Agent Financial Intelligence

Settles 8 of 14 compliance principles by arithmetic — a full screen in under 2ms with zero model calls, and 94 tests that pass with no API key and no network.

LangGraph

Constitutional AI

Python 3.12

AI TradeFlow

Shipped 

Inventory & Accounting for Wholesalers

Business-isolated books with a row-locked, idempotent posting path. 142 tests, including two-business leak tests and PostgreSQL oversell races. Credit-limit guard, void-by-reversal, WhatsApp collections desk, and Munshi AI with 5 read-only tools behind a deterministic constitution. Live no-signup demo; no production customer yet.

OpenAI Agents SDK

FastAPI

PostgreSQL

DevUnity Platform

Shipped 

This portfolio — and its own proof

A triage orchestrator over four specialists, a real MCP server on the official SDK, and a written constitution verified blocking 4 of 4 violations with no model reachable.

Agents SDK

FastMCP

Next.js 15

Bazaar

Shipped 

Unified B2B + B2C Marketplace

One codebase serving both retail checkout and a wholesale RFQ engine, with JazzCash, Easypaisa and card payments. PKR-native, three languages.

Next.js 15

FastAPI

Supabase

Textile ERP Platform

Module 1 live

Multi-tenant fabric mill ERP, Module 1 of 7 live

Roll register, yarn stock ledger, weaving/knitting, LC landed cost and roll traceability, isolated by PostgreSQL row-level security. 42 tests against real Postgres; live no-signup demo. CMT billing, ledgers and financials are spec'ed, not built.

FastAPI

PostgreSQL RLS

Next.js 15

Agent Factory (H5) [↗](#)

Shipped 

A General Agent that manufactures Custom Agents

Two-tier architecture where Claude Code manufactures OpenAI Agents SDK agents; SKILL.md as portable intelligence units. Platinum tier, deployed on Kubernetes + Dapr.

Claude Code

SKILL.md

Dapr

</> Key Projects

Textile ERP Platform (Pakistan)

In Development

2024 – Present

Founder

Full-scale ERP targeting Pakistan's textile and garment industry, from Fabric Mills to CMT units to garment exporters.

- › Fabric Mill Module (complete, live demo): roll/lot register with partial issuance, append-only yarn stock ledger, weaving & knitting sessions, LC imports with landed cost, roll traceability, 7 deterministic Mill Pulse alerts; tenant isolation via PostgreSQL row-level security, 42 tests
- › CMT Core (spec'ed, next): order lifecycle, auto-billing (4 bill types), BOM inventory, production sessions, dispatch
- › Financial (spec'ed): party ledgers, aging, P&L and cash flow for multi-city hubs (Faisalabad, Sialkot, Gujranwala, Karachi, Lahore)

Next.js 15

FastAPI

PostgreSQL RLS

SQLAlchemy 2

Vercel

[↗](#) Live Demo

Agent Factory (H5)

Panaversity Hackathon 2025

2025

Lead Developer

Two-tier agentic architecture: General Agent manufactures Custom Agents at scale.

- › General Agent (Claude Code) manufactures Custom Agent (OpenAI Agents SDK)
- › SKILL.md files as portable, reusable, monetizable intelligence units
- › Digital FTE pricing model targeting OpenAI Apps ecosystem (800M users)
- › Deployed on Kubernetes + Dapr

Claude Code

OpenAI SDK

MCP

SKILL.md

Dapr

Kubernetes

Professional Experience

Founder & CEO, CMT Stitching Operations

Texcot Embroidery Sourcing House

2020 – Present

Karachi, Pakistan

- › Engineered a multi-agent orchestration layer for financial-services research and Shari'ah screening, with separation of powers enforced structurally — no agent holds two of market data, weighting and verdict — rather than by prompt
- › Implemented MCP interoperability so agent capability is exposed as read-only tools on a protocol server, verified against a live client, instead of as bespoke HTTP endpoints
- › Architected deterministic guardrails from a written constitution, with a pattern layer that needs no model — verified blocking 4 of 4 violations with no model reachable, so enforcement survives a provider outage
- › Architected the Textile ERP Platform — order lifecycle, four auto-billing types, party ledgers and BOM inventory — to digitize Pakistan's textile value chain
- › Founded and operate a CMT stitching unit of 30–35 staff, running the full manufacturing lifecycle from sample fabric and design analysis through machine allocation, bulk production and inspection
- › Lead the business's digital marketing across social channels and lead-generation funnels

Marketing Manager

JK Embroidery

2016 – 2020

Karachi, Pakistan

- › Directed marketing and production planning for embroidery manufacturing operations
- › Engineered staff and equipment scheduling to clear production bottlenecks, implementing process improvements that held under peak order flow
- › Led a team of 30–35 staff, setting priorities against available resources and work-order flow

Country Manager / Buying Agent

Steven Berry

2003 – 2005

International

- › Managed and organized international shipments, providing clients guidance throughout procurement processes
- › Ensured products met established international quality standards across garment and textile categories
- › Conducted pre-production, in-production, and final inspection of apparel, accessories, and textile products



Digital Marketing Services

Full-spectrum digital marketing strategy for Dubai real estate, UAE construction, and Pakistani SMEs:

- › Social media management (Facebook, Instagram, X/Twitter, LinkedIn) + paid advertising campaigns
- › Property portal listings, lead generation funnels, and monthly performance reporting
- › Tailored strategies for GCC/expat/Pakistani diaspora investors; AED-priced service packages

Certifications & Credentials

9 completed courses and certificates from Cognizant & Anthropic. Every card links to the issuer's own record or the certificate itself.



CERTIFICATE OF COMPLETION

Agent Foundations

Agent Academy · Cognizant AI Lab

Completed September 30, 2026

[View certificate](#) 



VERIFIED COMPLETION BADGE

AI Fluency: Framework and foundations

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

[Verify with issuer](#) 



VERIFIED COMPLETION BADGE

Introduction to Claude Cowork

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

[Verify with issuer](#) 



VERIFIED COMPLETION BADGE

Claude Code 101

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

[Verify with issuer](#) 



VERIFIED COMPLETION BADGE

AI Fluency for builders

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

[Verify with issuer](#) 



VERIFIED COMPLETION BADGE

AI capabilities and limitations

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

[Verify with issuer](#) 



VERIFIED COMPLETION BADGE

Claude 101

Claude Academy · Anthropic's learning platform



VERIFIED COMPLETION BADGE

Claude Code in action

Claude Academy · Anthropic's learning platform

Issued September 18, 2026

[Verify with issuer](#) 

[Verify with issuer](#) 



VERIFIED COMPLETION BADGE

Introduction to Model Context Protocol

Claude Academy · Anthropic's
learning platform

[Verify with issuer](#) 



Education & Certifications

Governor Sindh Initiative for Artificial Intelligence & Computing (Panaversity)Feb 2024 – Present

Advanced study in Agentic AI and Generative AI

Forward Deployed Engineer Training — Track B (Accelerated), 13 weeks:
governed Vertical Systems of Record, stateless MCP (rev. 2026-07-28), multi-agent research systems, and structured extraction pipelines with validation and human review.

- TypeScript / JavaScriptReact / Next.jsPython
- Agent SDKs & AI frameworksLoop / Harness / Graph architectures

EXPECTED

Sindh Board / University of Karachi — Programme certification
Panaversity — PCAR-F — Architect Foundations, aligned to Anthropic's CCAR-F blueprint
Anthropic — Claude Certified Associate – Foundations → Architect – Foundations

Alim (5-Year Islamic Studies Course)
Burooj Institute, Karachi

Associate Degree in Textile Technology1997

Textile Institute of Pakistan (APTMA)

H.S.C Pre-Engineering1994

Pakistan Shipowner's Government College, Karachi

S.S.C Science1992

Ladybird Grammar School, Karachi